



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to SREERAM AGENCY

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

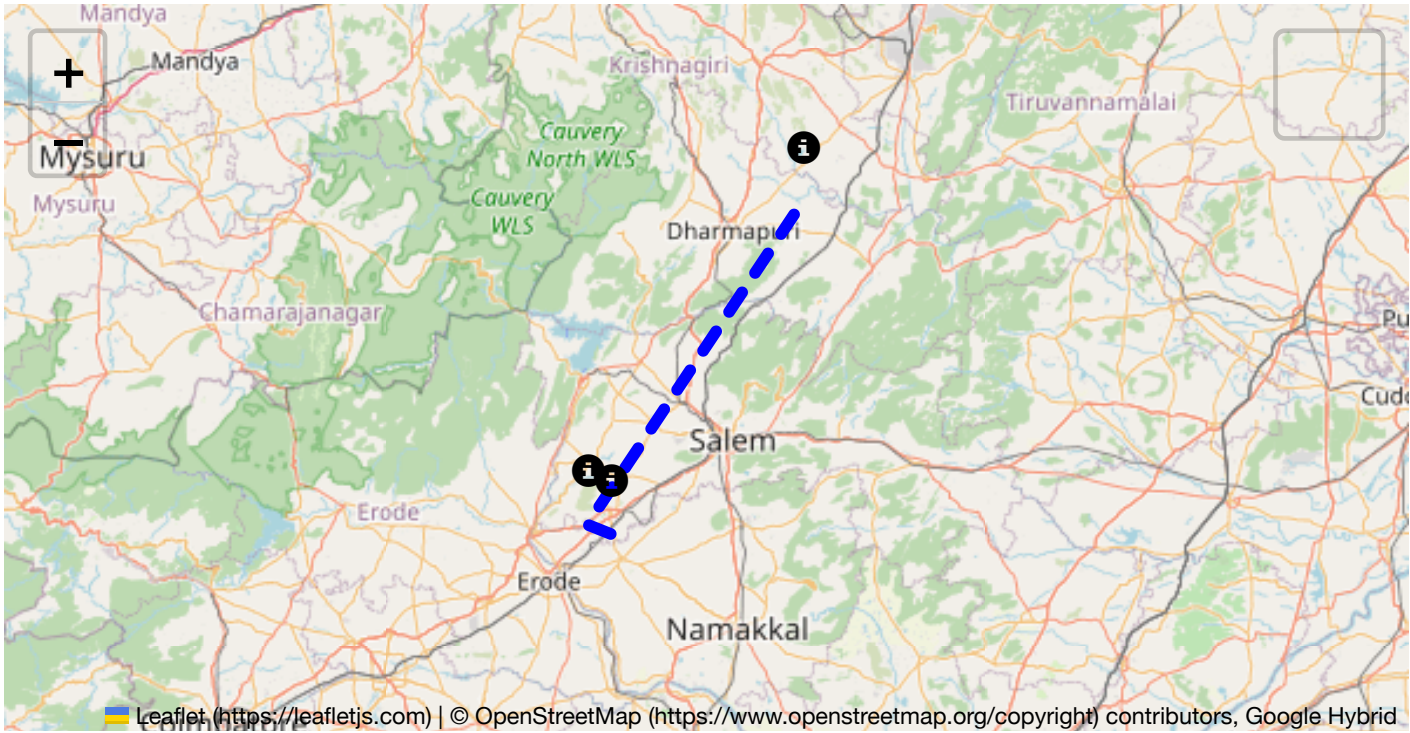
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

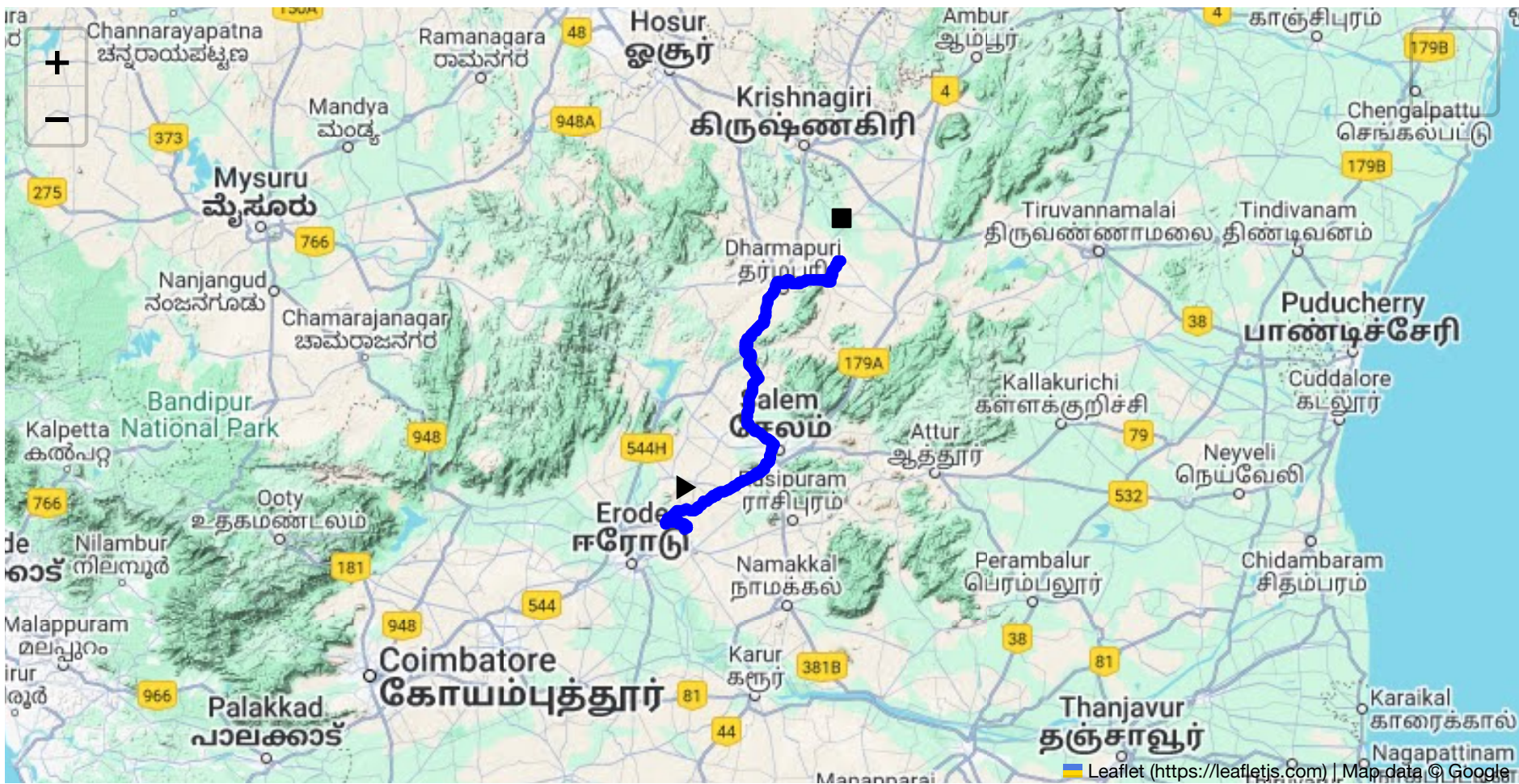
Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 142.30 km
Estimated Duration: 2.7 hours
Adjusted Duration (Heavy Vehicle): 3.4 hours

Start: (11.4381, 77.8734)
End: (12.201903, 78.322043)



Welcome to the Journey Risk Management Study

Route Safety Analysis: Sangagiri to Boomisamuthiram via Bhavani Main Road

1. Overview of the Route Map

The route traverses approximately 142.3 kilometers from Sangagiri, Tamil Nadu to Boomisamuthiram, passing through Bhavani Main Road. This journey involves rural and semi-urban roadways, with a mix of highways and local roads.

2. Typical Weather Conditions and Hazards

The region generally experiences a tropical climate, with hot summers, a monsoon season from June to September, and mild winters. Potential weather-related hazards include:

- **Monsoon:** Heavy rains leading to waterlogging and slippery roads.
- **Summer:** High temperatures that can affect vehicle performance, particularly heavy vehicles.

3. Traffic Patterns and Congestion

This route crosses a mix of high-traffic and low-traffic areas:

- **Peak Hours (Morning: 8-10 am, Evening: 5-7 pm):** Typically busy around urban centers, such as near major junctions or town entries.
- **Congestion-prone Areas:** Bhavani Main Road is frequently used by commuters, thus prone to congestion during peak hours.

4. Road Quality and Infrastructure

- **Highways:** Generally well-maintained, but watch for occasional potholes, especially post-monsoon.
- **Rural Roads:** May have uneven surfaces and narrower passages.

5. Suggestions for Alternative Routes

In the event of an emergency or unforeseen blockade:

- **Alternate Route 1:** Consider detouring via NH 544 for better infrastructure and emergency services.
- **Alternate Route 2:** Use local roads connecting to NH 44 for quicker access to medical facilities.

6. Local Regulations on Hazardous Material Transport

- **Permits Required:** Ensure all necessary permits for transporting hazardous materials are acquired.
- **Movement Restrictions:** Avoid transit during extreme weather advisories.

7. Historical Incidents

No major historical incidents specific to heavy vehicles noted, although heavy rain occasionally causes minor landslides in hilly areas.

8. Environmental Considerations

- **Sensitive Areas:** Nearby forests and agricultural zones; minimize noise and avoid fuel spills.
- **Wildlife Crossings:** Be vigilant in marked areas.

9. Communication Coverage

- **Urban and Highway Segments:** Generally good coverage.
- **Rural Areas:** Expect potential dead zones, particularly in remote stretches.

10. Estimated Emergency Response Times

- **Urban Areas:** Typically 20-30 minutes due to proximity to healthcare and fire services.
- **Rural Areas:** Could extend to 45-60 minutes due to distance and road conditions.

12. Overall Risk Assessment Summary

- **Moderate Risk:** Given the route involves a mix of well-maintained highways and less predictable rural roads. Potential weather and traffic conditions could pose challenges.
- **Recommendations:** Regularly check weather forecasts, keep communication devices charged, adhere to speed limits, and maintain a respectful distance from vehicles carrying similar loads.

Conclusion

The route from Sangagiri to Boomisamuthiram is generally passable but requires caution due to variable weather conditions and traffic congestion in places. Proper planning and vigilant driving practices can mitigate many potential hazards.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	12.13658, 78.16171	15 KM/Hr
1	Roundabout	High	12.14589, 78.17156	15 KM/Hr
2	Turn	High	11.43968, 77.87345	15 KM/Hr
3	Turn	High	11.44029, 77.87544	15 KM/Hr
4	Turn	High	11.44900, 77.87406	15 KM/Hr
5	Turn	High	11.45353, 77.85697	15 KM/Hr
6	Turn	Medium	11.45781, 77.85203	30 KM/Hr
7	Turn	High	11.46307, 77.84941	15 KM/Hr
8	Turn	High	11.45542, 77.81725	15 KM/Hr
9	Turn	High	11.45605, 77.81683	15 KM/Hr
10	Turn	Medium	11.53573, 77.95794	30 KM/Hr
11	Turn	Medium	11.62791, 78.11538	30 KM/Hr
12	Turn	Medium	11.63056, 78.11726	30 KM/Hr
13	Turn	Medium	11.64183, 78.11810	30 KM/Hr
14	Turn	Medium	11.64338, 78.11998	30 KM/Hr
15	Turn	Medium	11.74068, 78.05047	30 KM/Hr
16	Turn	Medium	11.78039, 78.05792	30 KM/Hr
17	Turn	Medium	11.86753, 78.08331	30 KM/Hr
18	Turn	Medium	11.86981, 78.08387	30 KM/Hr
19	Turn	Medium	11.88041, 78.08143	30 KM/Hr
20	Turn	Medium	11.91302, 78.05856	30 KM/Hr
21	Turn	Medium	11.93245, 78.06456	30 KM/Hr
22	Turn	Medium	11.95625, 78.04748	30 KM/Hr
23	Turn	High	11.95824, 78.04981	15 KM/Hr
24	Turn	High	11.96040, 78.04947	15 KM/Hr
25	Turn	Medium	11.96320, 78.05139	30 KM/Hr
26	Turn	Medium	11.96401, 78.05134	30 KM/Hr
27	Turn	Medium	11.96463, 78.05077	30 KM/Hr
28	Turn	Medium	11.96504, 78.04975	30 KM/Hr
29	Turn	High	11.96597, 78.04920	15 KM/Hr
30	Turn	Medium	11.96679, 78.04942	30 KM/Hr
31	Turn	Medium	11.96804, 78.05080	30 KM/Hr
32	Turn	Medium	11.98504, 78.06287	30 KM/Hr
33	Turn	High	12.14045, 78.13962	15 KM/Hr
34	Turn	High	12.14497, 78.16507	15 KM/Hr
35	Turn	Medium	12.14556, 78.27753	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
36	Turn	Medium	12.14594, 78.27904	30 KM/Hr
37	Turn	Medium	12.14461, 78.28382	30 KM/Hr
38	Turn	High	12.13994, 78.29107	15 KM/Hr
39	Turn	Medium	12.14388, 78.29402	30 KM/Hr
40	Turn	Medium	12.16686, 78.30090	30 KM/Hr
41	Turn	Medium	12.19754, 78.31632	30 KM/Hr
42	Turn	Medium	12.19883, 78.31841	30 KM/Hr
43	Turn	High	12.20238, 78.31919	15 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
2	hospital	C. N. Hospital	11.55724, 78.0096	30 km/h	Medium
3	hospital	Vinayaka Mission's Homeopathic Medical College & Hospital	11.583617, 78.059145	30 km/h	Medium
4	hospital	Vinayaka mission Kirubananadah variyar medical college hospital	11.584984, 78.0618417	30 km/h	Medium
5	hospital	vims	11.5854239, 78.0639188	30 km/h	Medium
8	hospital	Vinayaka Mission’s Kirupananda Variyar Medical College	11.5977669, 78.0850992	30 km/h	Medium
11	hospital	Bharathi Hospital, Salem	11.6492647, 78.1204131	30 km/h	Medium
12	hospital	Sisu Hospital Inida Ltd	11.6505276, 78.1205003	30 km/h	Medium
13	hospital	Aishwarya Hospital	11.6716291, 78.1267807	30 km/h	Medium
14	hospital	C.M. Hospital	11.6747345, 78.1255034	30 km/h	Medium
16	hospital	Sri Gokulam Speciality Hospital	11.6788724, 78.1274258	30 km/h	Medium
17	pharmacy	Healthy Pharmacy	11.6762714, 78.127744	30 km/h	Medium
18	pharmacy	Thulasi Pharmacy	11.6761991, 78.1277377	30 km/h	Medium
19	hospital	Global Medical Center	11.6867049, 78.1186697	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
20	hospital	Manipal Hospital	11.703072, 78.1026612	30 km/h	Medium
22	pharmacy	Salem Medicals	11.7343192, 78.0566844	30 km/h	Medium
24	clinic	Manimuthu Skin Clinic	11.7521189, 78.0484058	30 km/h	Medium
25	hospital	St. Joseph's Speciality Hospital	11.761611, 78.0522344	30 km/h	Medium
28	clinic	JSA Clinic	11.8700619, 78.0844982	30 km/h	Medium
29	hospital	L.R. Hospital	11.8703354, 78.0844554	30 km/h	Medium
30	hospital	Thoppur,Government Hospital	11.9398824, 78.0551658	30 km/h	Medium
32	clinic	Bee Tee Clinic	12.138889, 78.1576748	30 km/h	Medium
33	clinic	Sheela Clinic	12.138725, 78.1573436	30 km/h	Medium
34	clinic	Suguna Clinic	12.1386752, 78.1575468	30 km/h	Medium
35	hospital	Vijaya Hospital	12.1399758, 78.1532115	30 km/h	Medium
36	hospital	Navina Hospital	12.1367521, 78.1610296	30 km/h	Medium
37	clinic	Bharani Neurologist	12.135969, 78.16255	30 km/h	Medium
38	hospital	Pragadeesh Hospital	12.135284, 78.1591867	30 km/h	Medium
39	hospital	Kamatchi Hospital	12.1350608, 78.1594841	30 km/h	Medium
40	clinic	Malar Siddha Clinic	12.1350485, 78.159226	30 km/h	Medium
41	hospital	Kamala Nehru Hospital	12.13404, 78.1633476	30 km/h	Medium
42	hospital	D.N.V Hospital	12.1351328, 78.162869	30 km/h	Medium
43	clinic	Ragavendra Physiotherapy Clinic	12.135569, 78.1635247	30 km/h	Medium
44	clinic	Govindaraj Clinic	12.1344914, 78.1642372	30 km/h	Medium
45	hospital	Sri Priya Hospital	12.135347, 78.164391	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium
6	school	Vinayaka Missions Annapoorani Matric School	11.5971662, 78.0839876	30 km/h	Medium
7	university	Vinayaka Missions Vinayaka Missions Research Foundation	11.5974056, 78.0843617	30 km/h	Medium
9	school	Dist Education Training	11.6133799, 78.1022409	30 km/h	Medium
10	school	DIET - Salem	11.6144158, 78.1025078	30 km/h	Medium
15	college	Sona College	11.6768, 78.1257052	30 km/h	Medium
21	university	Periyar University	11.7186758, 78.0771089	30 km/h	Medium
23	school	Fathima Hr Sec School	11.7410508, 78.0479272	30 km/h	Medium
26	school	John Britto Matriculation Higher Secondary School	11.7676907, 78.0513092	30 km/h	Medium
27	college	Dhirajlal Gandhi College of Technology	11.7968014, 78.0562095	30 km/h	Medium
31	college	Laxminarayana Polytechnic College	12.0421277, 78.103203	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout
Risk Level: High

Speed Limit: 15 KM/Hr
Coordinates: 12.13658, 78.16171



Risk Type: Roundabout
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.14589, 78.17156



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44029, 77.87544



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44900, 77.87406



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45353, 77.85697



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45781, 77.85203



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.46307, 77.84941



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45542, 77.81725



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45605, 77.81683



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.53573, 77.95794



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.62791, 78.11538



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.63056, 78.11726



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64338, 78.11998



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.74068, 78.05047



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.78039, 78.05792



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.86753, 78.08331



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.86981, 78.08387



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.88041, 78.08143



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.91302, 78.05856



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.93245, 78.06456



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.95625, 78.04748



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.95824, 78.04981



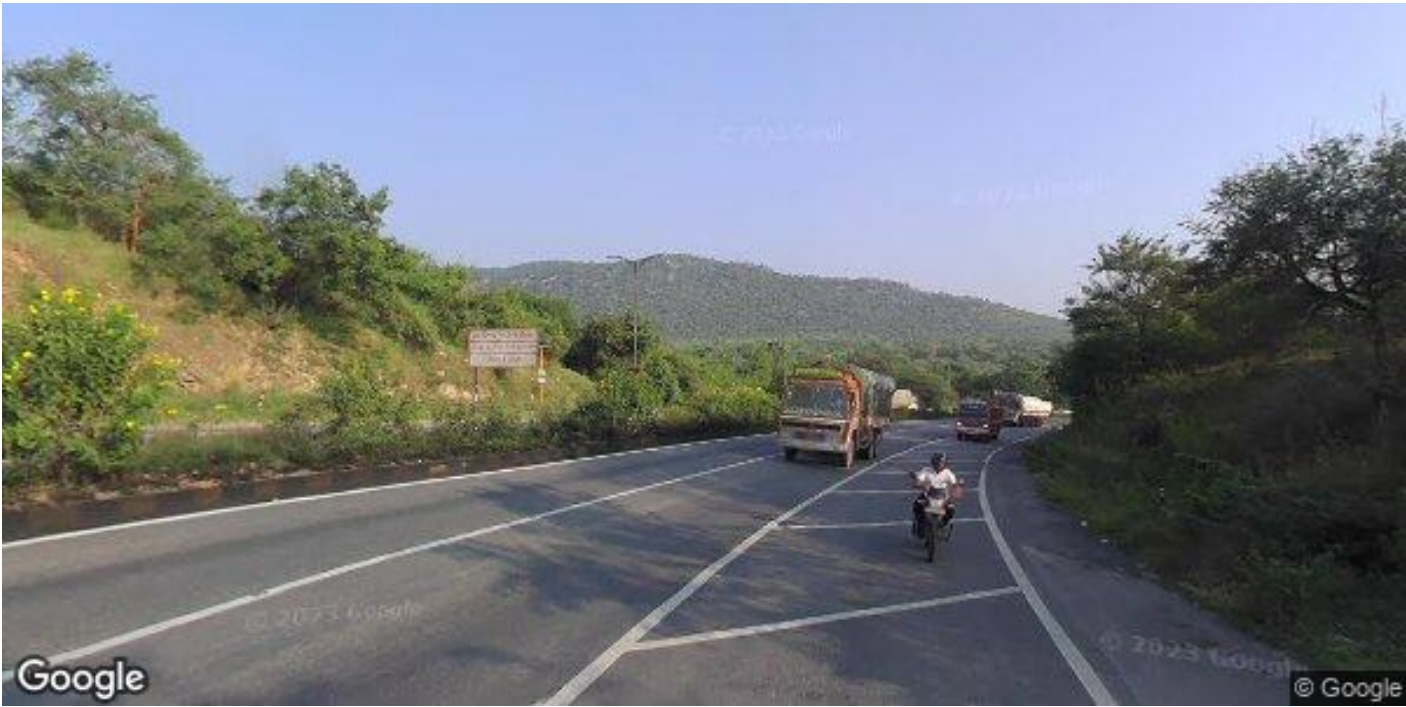
Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.96040, 78.04947



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96320, 78.05139



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96401, 78.05134



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96463, 78.05077



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96504, 78.04975



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.96597, 78.04920



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96679, 78.04942



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96804, 78.05080



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.98504, 78.06287



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.14045, 78.13962



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.14497, 78.16507



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.14556, 78.27753



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.14594, 78.27904



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.14461, 78.28382



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.13994, 78.29107



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.14388, 78.29402



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.16686, 78.30090



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.19754, 78.31632



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.19883, 78.31841



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.20238, 78.31919

